

**FECAP - Fundacao de Comercio Alvares Penteado**

**Entrega 2 - Disciplina de Algoritmos e Logica de Programacao**

# **Analise academica dos algoritmos presentes no jogo EcoCity**

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Projeto: EcoCity

Documento elaborado a partir do codigo da pasta src/Entrega 2/Frontend/EcoCity1/Assets/Scripts.

## 1. Introducao

O desenvolvimento de jogos digitais exige a aplicacao constante de estruturas fundamentais de programacao, como condicoes, repeticoes e estruturas de dados. No projeto EcoCity, essas construcoes aparecem tanto na logica principal da partida quanto no controle de interface, movimentacao de elementos, gerenciamento de estado e organizacao do tabuleiro.

Este trabalho apresenta uma analise academica das ocorrencias de if, else, for, while e array nos scripts do jogo, com o objetivo de demonstrar como esses elementos foram aplicados no desenvolvimento pratico do sistema.

## 2. Objetivo

Identificar, quantificar e explicar a utilizacao das estruturas if, else, for, while e array nos scripts do EcoCity, relacionando esses recursos com o funcionamento real do jogo.

## 3. Metodologia

A analise foi realizada exclusivamente sobre os scripts contidos em src/Entrega 2/Frontend/EcoCity1/Assets/Scripts. Foram desconsiderados arquivos externos ao codigo autoral do grupo, como exemplos do Unity, bibliotecas auxiliares, scripts de tutorial e pacotes de terceiros. A identificacao das estruturas foi feita por varredura textual dos arquivos C#, seguida da organizacao das ocorrencias por elemento, por arquivo e por linha.

## 4. Escopo analisado

```
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/Tile.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/CameraController.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/CityEventSystem.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/DiceVisual.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/EndingEvaluator.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/RoundManager.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/ScoreCalculator.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerStats.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/FinalResultScreen.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/PurchasePanel.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs
src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs
```

## 5. Resumo quantitativo

| Elemento | Total de ocorrencias | Arquivos com ocorrencias |
|----------|----------------------|--------------------------|
| if       | 402                  | 20                       |
| else     | 25                   | 9                        |
| for      | 27                   | 8                        |
| while    | 11                   | 5                        |
| array    | 38                   | 6                        |

## 6. Analise dos elementos

### Estruturas condicionais if

A estrutura if é utilizada para validar condições antes da execução de uma ação. No EcoCity, ela aparece com muita frequência para conferir estados do jogo, verificar referências nulas, impedir erros em tempo de execução e controlar comportamentos da interface e da lógica da partida.

Exemplo representativo retirado de: **src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/Tile.cs**

```
if (buyer == null || Data == null)
{
    return;
}
```

### Estruturas condicionais else

A estrutura else complementa as decisões condicionais, oferecendo um caminho alternativo quando a condição principal não é satisfeita. No jogo, esse recurso é importante para definir respostas padrão, tratamentos alternativos e configurações de fallback.

Exemplo representativo retirado de: **src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs**

```
if (backgroundSprite != null)
{
    background.sprite = backgroundSprite;
    background.color = backgroundTint;
    background.type = Image.Type.Simple;
    background.preserveAspect = false;
}
else
{
    background.sprite = null;
    background.color = backgroundColor;
    background.type = Image.Type.Simple;
}
```

### Estruturas de repetição for

A estrutura for é aplicada quando o sistema conhece previamente a quantidade de repetições. No projeto, ela é usada para percorrer listas, criar elementos do tabuleiro, reposicionar componentes de interface e distribuir objetos visuais no cenário.

Exemplo representativo retirado de: **src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs**

```
for (int index = 0; index < path.Count; index++)
{
    Vector2Int gridPosition = path[index];
    Vector3 worldPosition = GetWorldPosition(gridPosition.x, gridPosition.y);

    Tile tileInstance = Instantiate(tilePrefab, worldPosition, Quaternion.identity, transform);
    TileData tileData = GenerateTileData(index);

    tileInstance.Initialize(tileData, index);
    Tiles.Add(tileInstance);
}
```

### Estruturas de repetição while

A estrutura while aparece em processos que dependem de uma condicao de continuidade. No EcoCity, isso se relaciona principalmente com animacoes, esperas controladas, rotinas de movimento e verificacoes repetidas ao longo do tempo.

Exemplo                      representativo                      retirado                      de:                      **src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs**

```
while (elapsed < hopDuration)
{
    elapsed += Time.deltaTime;
    float t = Mathf.Clamp01(elapsed / hopDuration);
    float forwardT = EaseInOutCubic(t);
    float hopOffset = Mathf.Sin(t * Mathf.PI) * hopHeight;

    Vector3 flatPosition = Vector3.Lerp(startPosition, targetPosition, forwardT);
    desiredPosition = flatPosition + Vector3.up * hopOffset;
    float tiltAmount = Mathf.Sin(t * Mathf.PI) * forwardTiltAngle;
    Quaternion tiltedRotation = baseRotation * Quaternion.Euler(-tiltAmount, 0f, 0f);
    desiredRotation = Quaternion.Slerp(desiredRotation, tiltedRotation, 0.35f);
    transform.position = desiredPosition;
    transform.rotation = desiredRotation;
    OnMoveStep?.Invoke(desiredPosition);

    yield return null;
}
```

### Estruturas de armazenamento com array

O array e uma estrutura de armazenamento de dados em sequencia, com acesso por indice. No jogo, ele aparece tanto em declaracoes de colecoes fixas quanto no uso de vetores retornados por componentes do Unity, sendo relevante para organizacao de dados e manipulacao de elementos do sistema.

Exemplo                      representativo                      retirado                      de:                      **src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs**

```
TileType[] pathTypes =
{
    TileType.Start,
    TileType.Residential,
    TileType.Park,
    TileType.Factory,
    TileType.Shopping,
    TileType.School,
    TileType.Hospital,
    TileType.Factory,
    TileType.TreatmentPlant,
    TileType.Residential,
    TileType.Park,
    TileType.Shopping,
    TileType.SolarPlant,
    TileType.FoodCourt,
    TileType.Residential,
    TileType.Factory,
    TileType.School,
    TileType.Park,
    TileType.Hospital,
    TileType.Shopping
};
```

## 7. Conclusao

A analise dos scripts do EcoCity demonstra que as estruturas estudadas na disciplina estao presentes de maneira concreta e funcional no desenvolvimento do jogo. O if e o else aparecem fortemente ligados ao controle de fluxo e validacao de estados; o for e o while atuam em rotinas de repeticao para construcao e animacao; e os arrays participam da organizacao de dados e do acesso sequencial a informacoes do sistema.

Dessa forma, o projeto evidencia a aplicacao pratica dos fundamentos de algoritmos e logica de programacao em um contexto real de desenvolvimento de software interativo.

## 8. Apendice - Ocorrencias completas

As tabelas a seguir apresentam todas as ocorrencias identificadas nos scripts analisados, organizadas por elemento, arquivo e linha.

### Estruturas condicionais if

Total de ocorrencias: **402**.

| Arquivo                                                                         | Ocorrencias |
|---------------------------------------------------------------------------------|-------------|
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | 11          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/Tile.cs                    | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/CameraController.cs       | 11          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | 33          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/CityEventSystem.cs          | 5           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/DiceVisual.cs               | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/EndingEvaluator.cs          | 6           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | 3           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs           | 65          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs             | 26          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/RoundManager.cs             | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/ScoreCalculator.cs          | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs            | 9           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerStats.cs            | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/FinalResultScreen.cs          | 5           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    | 77          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/PurchasePanel.cs              | 26          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | 71          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | 22          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | 24          |

| Linha | Arquivo                                                              | Trecho                                      |
|-------|----------------------------------------------------------------------|---------------------------------------------|
| 21    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.Starts                     |
| 29    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.index >= Tiles.Count)      |
| 42    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.Cs                         |
| 47    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.isPlaying)                 |
| 59    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.null)                      |
| 80    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.ExpectedTileCount)         |
| 93    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.Cs                         |
| 111   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.isPlaying)                 |
| 127   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.visual != null)            |
| 134   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.visual != null)            |
| 201   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (BoardManager.index >= pathTypes.Length) |
| 30    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs | if (Data == null    Data == null)           |

| Linha | Arquivo                                                                    | Trecho                                                                     |
|-------|----------------------------------------------------------------------------|----------------------------------------------------------------------------|
| 46    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 70    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | CameraState == null)                                                       |
| 75    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | CameraState == CameraState.Idle)                                           |
| 97    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | CameraState == CameraState.Idle)                                           |
| 116   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | CameraState == CameraState.Idle)                                           |
| 122   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | CameraState == CameraState.Focus && focusTarget.HasValue)                  |
| 128   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | CameraState == CameraState.Cinematic)                                      |
| 183   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 217   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 225   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter.magnitude < 0.001f)                                            |
| 259   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 263   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 101   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 106   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 111   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 126   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 131   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 136   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 147   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 154   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/DetailCamera.cs      | BoardCenter == null)                                                       |
| 172   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 180   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu.Contains("play"))                                                 |
| 215   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 234   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu.Background != null)                                               |
| 255   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 264   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu != null)                                                          |
| 279   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 300   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 377   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 392   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 431   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu != null)                                                          |
| 436   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu != null)                                                          |
| 451   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 458   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == "PanelTitle"    objectName == "PanelBody")                     |
| 463   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu.GetComponent<Button>(true) != null)                               |
| 479   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 498   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 503   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu != null)                                                          |
| 525   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 540   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu != null && currentTransform.name == objectName)                   |
| 556   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 560   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 579   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 583   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == null)                                                          |
| 628   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenu.cs            | MainMenu == scenePath.GetFileNameWithoutExtension(scenePath) == sceneName) |
| 44    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/CityFiveStatsSystem.cs | CityFiveStatsSystem == null)                                               |
| 66    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/CityFiveStatsSystem.cs | CityFiveStatsSystem == null)                                               |
| 104   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/CityFiveStatsSystem.cs | CityFiveStatsSystem.Pollution >= 70)                                       |
| 116   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/CityFiveStatsSystem.cs | CityFiveStatsSystem.WellBeing <= 35)                                       |



| Linha | Arquivo                                                                       | Trecho                                             |
|-------|-------------------------------------------------------------------------------|----------------------------------------------------|
| 137   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameStateSystem.cs        | result.Triggered)                                  |
| 12    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/DiceVisualForm.cs         | == null)                                           |
| 45    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StrategicEvaluator.cs     |                                                    |
| 58    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StrategicEvaluator.cs     | >= 75    stats.WellBeing <= 25    score < 700f)    |
| 69    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StrategicEvaluator.cs     | && (stats.Pollution >= 55    stats.WellBeing       |
| 80    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StrategicEvaluator.cs     | >= 75 && stats.Pollution <= 30 && stats.Money      |
| 91    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StrategicEvaluator.cs     | && stats.WellBeing >= 55 && stats.Pollution <= 45  |
| 116   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StrategicEvaluator.cs     |                                                    |
| 20    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameUIScrollController.cs |                                                    |
| 25    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameUIScrollController.cs |                                                    |
| 107   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameUIScrollController.cs | clip == null)                                      |
| 47    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null    playerPiece == null    playerStats == n |
| 64    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | EnsureTilesReady())                                |
| 85    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null)                                           |
| 91    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null)                                           |
| 103   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameScreenShower.cs       |                                                    |
| 109   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameScreenShower.cs       | KeyCode.Escape))                                   |
| 116   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameScreenShower.cs       |                                                    |
| 121   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameScreenShower.cs       | KeyCode.Space) && !playerPiece.IsMoving && !is     |
| 132   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null && !roundManager.CanRoll)                  |
| 137   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null)                                           |
| 156   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | <= 0)                                              |
| 170   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null)                                           |
| 185   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | && !finalScreenShown)                              |
| 199   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null)                                           |
| 204   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | == null)                                           |
| 209   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            | && playerPiece != null)                            |
| 214   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            |                                                    |
| 219   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs            |                                                    |
| 223   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 229   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 233   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 239   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 244   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 249   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 254   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 259   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 264   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 268   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 275   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 279   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 286   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 290   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 294   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 323   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 334   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 350   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null)                                           |
| 374   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | == null    playerPiece == null)                    |
| 379   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameHUD.cs                | EnsureTilesReady())                                |

| Linha | Arquivo                                                               | Trecho                                                                                        |
|-------|-----------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| 399   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 406   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (tile.IsPurchased())                                                                       |
| 408   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (playerStats.Money == null)                                                                |
| 436   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (tile.IsPurchased() && playerStats.Money < previousMoney)                                  |
| 464   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (tile.IsPurchased() && currentTile.owner != null && currentTile.owner != playerStats)      |
| 466   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (playerStats.Money == null)                                                                |
| 492   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (tile.IsPurchased() && tile.IsOwnedByPlayer)                                               |
| 498   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (startEvent != null && startEvent.Triggered)                                               |
| 518   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (data.WellBeingImpact == 0 && data.EnvironmentalImpact == 0 && data.PopulationImpact == 0) |
| 538   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 551   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 564   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 581   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (GameManager.Instance.TrainadsScreenShown)                                                 |
| 592   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (playerStats == null    playerStats.Money == null)                                         |
| 601   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (tile.IsPurchased()    tile.IsOwnedByPlayer)                                               |
| 609   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (playerStats.Money != 0)                                                                   |
| 616   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (GameManager.Instance.TrainadsScreenShown)                                                 |
| 620   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (roundEvent != null && roundEvent.Triggered)                                               |
| 642   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 677   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (GameManager.Instance.TrainadsScreenShown)                                                 |
| 684   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 726   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 731   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (purchasePanel != null && purchasePanel.IsVisible)                                         |
| 738   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 757   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (GameManager.Instance.TrainadsScreenShown)                                                 |
| 807   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs |                                                                                               |
| 904   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs | if (roundManager == null)                                                                     |
| 71    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 96    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 101   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (Input.GetKeyDown(KeyCode.Space)    Input.GetKeyDown(KeyCode.F))                           |
| 106   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 114   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 137   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 142   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 149   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 156   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 163   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 170   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 191   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (Input.GetKeyDown(KeyCode.Space))                                                          |
| 202   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 263   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (cameraTransform != null)                                                                  |
| 268   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 301   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 313   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 326   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 350   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (isCharging)                                                                               |
| 417   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (chargeAudio.isPlaying)                                                                    |
| 428   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   |                                                                                               |
| 435   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs   | if (chargeAudio.isPlaying)                                                                    |

| Linha | Arquivo                                                                     | Trecho                          |
|-------|-----------------------------------------------------------------------------|---------------------------------|
| 462   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs         | GamePhysics.Dice != null)       |
| 470   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs         | GamePhysics.Dice != null)       |
| 478   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs         | GamePhysics.Dice.cs             |
| 483   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/Physics/Dice.cs         | GamePhysics.Dice.cs             |
| 32    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameManager.cs          | GameManager.cs                  |
| 40    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/RandomManager.cs        | maxRounds)                      |
| 5     | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StateCalculator.cs      | StateCalculator.cs              |
| 15    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/StateCalculator.cs      | StateCalculator.cs              |
| 27    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiecesStatic    | PlayerPiecesStatic              |
| 42    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiecesTransform | PlayerPiecesTransform           |
| 58    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.cs                 |
| 64    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.EnsureTilesReady() |
| 82    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.cs                 |
| 87    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.cs                 |
| 93    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.EnsureTilesReady() |
| 154   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.cs                 |
| 168   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.cs                 |
| 59    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPieces          | PlayerPieces.cs                 |
| 53    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/ResultsScreen.cs          | ResultsScreen.cs                |
| 86    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/ResultsScreen.cs          | ResultsScreen.cs                |
| 125   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/ResultsScreen.cs          | ResultsScreen.cs                |
| 195   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/ResultsScreen.cs          | ResultsScreen.cs                |
| 293   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/ResultsScreen.cs          | ResultsScreen.cs                |
| 56    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 62    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 69    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 74    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 79    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 84    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 98    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 103   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 114   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 125   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 130   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 135   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 140   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 145   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 159   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 177   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 182   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 187   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 192   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 203   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 214   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 225   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 238   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 244   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 257   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |
| 271   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                | GameHUD.cs                      |

| Linha | Arquivo                                                                                                | Trecho |
|-------|--------------------------------------------------------------------------------------------------------|--------|
| 284   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/ChargePanel == null    powerSlider == null)  |        |
| 290   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/FindImage != null)                           |        |
| 294   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Power < 0.5f)                                |        |
| 306   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/InstructionsLabel != null)                   |        |
| 308   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Power == 0f)                                 |        |
| 326   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD > 0f)                                        |        |
| 342   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD == null)                                     |        |
| 355   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/ChargePanel == null)                         |        |
| 359   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/PanelTransform != null)                      |        |
| 365   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Slider == null)                              |        |
| 369   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/SliderTransform != null)                     |        |
| 375   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Label == null)                               |        |
| 379   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/LabelTransform != null)                      |        |
| 385   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/InstructionsLabel == null)                   |        |
| 389   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/InstructionsTransform != null)               |        |
| 395   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/FindImage == null && powerSlider != null)    |        |
| 399   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/PanelTransform != null)                      |        |
| 405   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Text == null)                                |        |
| 410   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/FindRoot == null)                            |        |
| 414   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/PanelTransform != null)                      |        |
| 420   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/FindStatsRoot == null)                       |        |
| 424   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/PanelLeftTransform != null)                  |        |
| 430   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/MessagePanelRoot == null)                    |        |
| 434   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/MessagePanelTransform != null)               |        |
| 440   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/MessagePanelRoot == null)                    |        |
| 444   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/PurchasePanelTransform != null)              |        |
| 450   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/ToastRoot == null)                           |        |
| 454   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/EventToastTransform != null)                 |        |
| 460   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/AdPanelRoot == null)                         |        |
| 464   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/TutorialTransform != null)                   |        |
| 470   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/LoadingText == null)                         |        |
| 475   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/LocationText == null)                        |        |
| 480   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Text == null)                                |        |
| 485   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Text == null)                                |        |
| 490   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/ResultText == null)                          |        |
| 495   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/MessageText == null)                         |        |
| 500   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/AdText == null)                              |        |
| 505   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/ToastText == null)                           |        |
| 516   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/MissingPowerReferences)                      |        |
| 541   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Canvas != null)                              |        |
| 554   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs == null)                                  |        |
| 559   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD..name == childName)                          |        |
| 569   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Result != null)                              |        |
| 583   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Text != null)                                |        |
| 590   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Canvas == null)                              |        |
| 617   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/AdPanelRoot != null && tutorialText != null) |        |
| 624   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/Canvas == null)                              |        |
| 679   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/AdPanelRoot == null    tutorialText == null) |        |
| 701   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD/ToastRoot != null && eventToastText != null) |        |

| Linha | Arquivo                                                                               | Trecho                              |
|-------|---------------------------------------------------------------------------------------|-------------------------------------|
| 708   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/Canvas == null)            |                                     |
| 750   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/Root == null)              |                                     |
| 43    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 49    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 55    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 66    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               | tile.Data == null    stats == null) |
| 79    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 84    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 89    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 94    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 101   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 109   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 114   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 119   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               | && feedbackText != null)            |
| 125   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 136   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               | tile.Data == null    stats == null) |
| 156   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 162   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 169   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 182   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               | playerStats == null)                |
| 204   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               | && feedbackText != null)            |
| 211   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 224   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 233   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 242   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 265   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 270   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 281   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UF/UFGameUI/RootPanel.cs               |                                     |
| 99    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 113   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | == 0)                               |
| 196   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | && x < clearMaxX)                   |
| 210   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | && z < clearMaxZ)                   |
| 254   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 265   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | districtLevel))                     |
| 286   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 291   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 307   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 312   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 317   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 350   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | theme == DistrictTheme.Residential) |
| 383   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 425   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 434   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 451   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 484   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | && x < clearMaxX)                   |
| 491   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs | && z < clearMaxZ)                   |
| 522   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 526   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |
| 532   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/Board/BoardEnvironmentVisual.cs |                                     |



| Linha | Arquivo                                                                                                             | Trecho                                                |
|-------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| 534   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentPrefabs.cs                                    | (Random.value < 0.5f))                                |
| 538   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentPrefabs.cs                                    | (!m.isInPoolOfObjectsToDestroy))                      |
| 546   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadBuildingPrefabs.cs                                       | (!m.isInPoolOfObjectsToDestroy))                      |
| 552   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(prefabEnvironmentVisual.cs                                       |                                                       |
| 556   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 561   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadBuildingPrefabs.cs                                       | (!m.isInPoolOfObjectsToDestroy))                      |
| 567   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allStrands.environment2 && Random.value > 0.35f)                 |                                                       |
| 571   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentPrefabs.cs                                    |                                                       |
| 577   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allStrands.environment)Visual.cs                                 |                                                       |
| 581   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 589   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 594   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadBuildingPrefabsVisual.cs                                 |                                                       |
| 622   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(prefabEnvironmentPrefabcs.Length == 0)                           |                                                       |
| 632   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 639   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (keywords.Length == 0)                                |
| 648   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentContains(keyword))                            |                                                       |
| 656   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 661   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 674   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 688   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 727   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (currentLight.type == LightType.Directional))         |
| 736   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 806   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 863   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (currentLight.type == LightType.Directional))         |
| 879   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 884   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 910   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 918   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 933   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs    streetDetailPrefabs.Length == 0) |                                                       |
| 942   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 949   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (prefabName.Contains("street"))                       |
| 957   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 970   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 994   | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1039  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(prefabEnvironmentPrefabcs.Length == 0)                           |                                                       |
| 1048  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     |                                                       |
| 1054  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1067  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1074  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     | (bounds.size.y <= 0.001f    bounds.size.x <= 0.001f)) |
| 1084  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1094  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1108  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1144  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1162  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (transform.localScale = Vector3.one)                  |
| 1167  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (transform.localScale = Vector3.one)                  |
| 1172  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Visual/BroadEnvironmentVisual.cs                                     | (transform.localScale = Vector3.one)                  |
| 1196  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |
| 1209  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     | (transform.localScale = Vector3.one)                  |
| 1216  | src/Entrega_2/Frontend/EcoCity1/Assets/Scripts/Ufs(allBroadEnvironmentVisual.cs                                     |                                                       |

| Linha | Arquivo                                                                         | Trecho                                                                                             |
|-------|---------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1227  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | VisualBoardEnvironmentVisual.cs                                                                    |
| 58    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs    buildingPurchased)                                                     |
| 71    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs    boardCenter == null)                                                   |
| 76    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs    tile.IsPurchased)                                                      |
| 90    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 98    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 106   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 125   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs    currentTile == null)                                                   |
| 139   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 152   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 177   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs    currentTile == null)                                                   |
| 211   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 248   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 279   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 284   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs && prefabTreatmentPlant == null)                                          |
| 291   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 304   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 391   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 418   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 549   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 575   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 582   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 591   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | VisualPropertySpawner.cs                                                                           |
| 47    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 67    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 82    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 89    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 96    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 110   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 115   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs && currentTileIndex < allTiles.Count)                                   |
| 123   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs    matHighlight == null)                                                |
| 137   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs >= allTiles.Count)                                                      |
| 145   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 150   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 157   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 170   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 211   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 218   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 223   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs && purchasedMaterials.TryGetValue(index, out Material originalMaterial) |
| 229   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 240   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 253   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 262   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 275   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 277   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 352   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs                                                                         |
| 359   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | VisualTileVisualManager.cs && material != null)                                                    |

## Estruturas condicionais else

Total de ocorrencias: 25.

| Arquivo                                                                         | Ocorrencias |
|---------------------------------------------------------------------------------|-------------|
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/CameraController.cs       | 3           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | 5           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs           | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/FinalResultScreen.cs          | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    | 4           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | 6           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      | 2           |

| Linha | Arquivo                                                                         | Trecho                                                        |
|-------|---------------------------------------------------------------------------------|---------------------------------------------------------------|
| 115   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            |                                                               |
| 80    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/CameraController.cs       | else if (cameraController == CameraState.Follow)              |
| 84    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/CameraController.cs       | else if (cameraController == CameraState.Focus)               |
| 89    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Camera/CameraController.cs       |                                                               |
| 185   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | if (lowerName.Contains("option"))                             |
| 190   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | if (lowerName.Contains("credit"))                             |
| 195   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | if (lowerName.Contains("exit")    lowerName.Contains("quit")) |
| 200   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | if (lowerName.Contains("back"))                               |
| 510   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               |                                                               |
| 451   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs           |                                                               |
| 811   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs           |                                                               |
| 199   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/FinalResultScreen.cs          |                                                               |
| 298   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    |                                                               |
| 312   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    | if (power < 0.4f)                                             |
| 316   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    | if (power < 0.75f)                                            |
| 320   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    |                                                               |
| 269   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                               |
| 355   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                               |
| 744   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                               |
| 814   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                               |
| 922   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                               |
| 1176  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                               |
| 423   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        |                                                               |
| 162   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      |                                                               |
| 281   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/TileVisualManager.cs      |                                                               |



## Estruturas de repeticao for

Total de ocorrencias: 27.

| Arquivo                                                                         | Ocorrencias |
|---------------------------------------------------------------------------------|-------------|
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | 6           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs            | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/FinalResultScreen.cs          | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | 13          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | 1           |

| Linha | Arquivo                                                                         | Trecho                                           |
|-------|---------------------------------------------------------------------------------|--------------------------------------------------|
| 68    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | for (int index = 0; index < path.Count; index++) |
| 107   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | transform.childCount - 1; childIndex >= 0; ch    |
| 147   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | < gridSize; x++)                                 |
| 152   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | < gridSize; z++)                                 |
| 157   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | gridSize - 2; x >= 0; x--)                       |
| 162   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | gridSize - 2; z > 0; z--)                        |
| 365   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | index < buttons.Length; index++)                 |
| 624   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | index < SceneManager.sceneCountInBuildSettings:  |
| 125   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | noteIndex < frequencies.Length; noteIndex++      |
| 129   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | sampleIndex < samplesPerNote; sampleI            |
| 101   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs            | movedSteps < steps; movedSteps++)                |
| 191   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/FinalResultScreen.cs          | transform.childCount - 1; childIndex >= 0; ch    |
| 564   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/UI/GameHUD.cs                    | index = 0; index < parent.childCount; index++)   |
| 194   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | cityMaxX; x += blockSize + roadWidth)            |
| 208   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | cityMaxZ; z += blockSize + roadWidth)            |
| 240   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | roadWidth; startX < cityMaxX - block             |
| 242   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | roadWidth; startZ < cityMaxZ - b                 |
| 482   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | cityMaxX; x += blockSize + roadWidth)            |
| 489   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | cityMaxZ; z += blockSize + roadWi                |
| 717   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | index < landmarkPositions.Length; index++)       |
| 760   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | band; band++                                     |
| 785   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | index < buildingsPerSide; index++)               |
| 829   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | side; side++                                     |
| 836   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | index < treeGroupsPerSide; index++)              |
| 845   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | treeCount; tree++)                               |
| 1115  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | index < renderers.Length; index++)               |
| 329   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | index < localPositions.Length; index++)          |

## Estruturas de repeticao while

Total de ocorrencias: 11.

| Arquivo                                                                  | Ocorrencias |
|--------------------------------------------------------------------------|-------------|
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/DiceVisual.cs        | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs    | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs      | 4           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs     | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs | 3           |

| Linha | Arquivo                                                                  | Trecho                                              |
|-------|--------------------------------------------------------------------------|-----------------------------------------------------|
| 20    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/DiceVisual.cs        | while (elapsedTime < rollDuration)                  |
| 431   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs    | while (diceIsOnMa)                                  |
| 480   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameController.cs    | while (diceIsOnMa)                                  |
| 309   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs      | Raycast(transform.position, Vector3.down, out hitIn |
| 321   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs      | if (distance > 0.7f)                                |
| 379   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs      | moveToCameraDuration)                               |
| 396   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs      | returnDuration)                                     |
| 113   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Player/PlayerPiece.cs     | while (elapsed.es hopDuration)                      |
| 159   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs | while (elapsedDuration)                             |
| 187   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs | while (elapsedDuration)                             |
| 219   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs | while (elapsedDuration)                             |

## Estruturas de armazenamento com array

Total de ocorrências: **38**.

| Arquivo                                                                         | Ocorrências |
|---------------------------------------------------------------------------------|-------------|
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | 1           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Core/MainMenuUI.cs               | 4           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | 9           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/PhysicalDice.cs             | 2           |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | 19          |
| src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        | 3           |

| Linha | Arquivo                                                                         | Trecho                                                                                     |
|-------|---------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| 177   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | Board.GetComponent<BoardManager>().pass =                                                  |
| 168   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | GetComponentInChildren<Button>(true);                                                      |
| 363   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | panelTransform.GetComponentInChildren<Button>(t                                            |
| 447   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | GetComponentInChildren<Text>().text = panelTransform.GetComponentInChildren<Text>().text;  |
| 536   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Board/BoardManager.cs            | GetComponentInChildren<Transform>().transform = Resources.FindObjectsOfTypeAll<Transform>( |
| 34    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 540f, 680f }, 0.08f);                                  |
| 35    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 620f, 820f, 980f }, 0.07f);                            |
| 36    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 320f, 250f }, 0.09f);                                  |
| 37    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 420f, 360f, 300f }, 0.07f);                            |
| 38    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 700f, 860f, 1020f }, 0.07f);                           |
| 39    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 480f, 390f, 310f }, 0.07f);                            |
| 40    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(new[] { 760f, 640f, 840f, 520f }, 0.07f);                      |
| 118   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(float[] frequencies, float noteDur                             |
| 123   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(float[] frequencies, float noteDur                             |
| 50    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(float[] frequencies, float noteDur                             |
| 51    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Game/GameAudioController.cs      | CreateAudioClipFromSequence(float[] frequencies, float noteDur                             |
| 22    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | lowRiseBuildingPrefabs;                                                                    |
| 23    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | midRiseBuildingPrefabs;                                                                    |
| 24    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | tallRiseBuildingPrefabs;                                                                   |
| 25    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | treePrefabs;                                                                               |
| 26    | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | streetDetailPrefabs;                                                                       |
| 520   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | SetThemeKeywords(theme);                                                                   |
| 605   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | SetThemeKeywords(DistrictTheme theme)                                                      |
| 609   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | new[] { "house", "res", "home", "c                                                         |
| 610   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | new[] { "office", "shop", "store",                                                         |
| 611   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | new[] { "city", "hall", "school", "hosp                                                    |
| 612   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | new[] { "eco", "green", "solar", "park", "                                                 |
| 620   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | ChoosePrefabFromTheme(GameObject[] prefabs, string[] k                                     |
| 701   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                                                            |
| 709   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs |                                                                                            |
| 859   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | FindObjectsByType<Light>(FindObjectsInactive.Exclu                                         |
| 1037  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | ChoosePrefabFromTheme(GameObject[] prefabs)                                                |
| 1106  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | GetComponentInChildren<Renderer>(true)                                                     |
| 1158  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | GetComponentInChildren<Transform>(true);                                                   |
| 1232  | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/BoardEnvironmentVisual.cs | GetComponentInChildren<Collider>(true)                                                     |
| 320   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertySpawner.cs        |                                                                                            |

| Linha | Arquivo                                                                 | Trecho                                             |
|-------|-------------------------------------------------------------------------|----------------------------------------------------|
| 327   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertyDrawer.cs | float f, 0f, 8f };                                 |
| 412   | src/Entrega 2/Frontend/EcoCity1/Assets/Scripts/Visual/PropertyDrawer.cs | RenderTarget.GetComponentsInChildren<Renderer>()); |